

Data Used, Synthetic Data, And EDA

This file explains the data used in the project, what is real, what is synthetic, and what the EDA shows.

Data Sources

Real Source Data

The real source dataset is the NASA POWER meteorological CSV:

```
data/raw/Dataset_5yrs.csv
```

After cleaning:

```
data/processed/cleaned_nasa_power.csv
```

The dataset contains hourly weather/solar resource values:

Field	Meaning
GHI	Global horizontal irradiance.
DNI	Direct normal irradiance.
Diffuse irradiance	Diffuse solar component.
Temperature	Ambient temperature.
Wind speed	Wind speed at 10 m.
Humidity	Relative humidity.
Precipitation	Corrected precipitation.

Current cleaned dataset:

Metric	Value
Rows	52,608
Start timestamp	2020-01-01 05:30:00
End timestamp	2026-01-01 04:30:00
Duplicate timestamps	0
Missing cells after cleaning	0

Synthetic Data

NASA POWER does not provide:

- Facility load demand.
- Grid tariff.
- Battery SoC.
- Battery power.
- Grid import.
- Dispatch action.

So the project generates these under realistic constraints.

Synthetic dataset:

```
data/processed/ems_dataset.csv
```

What Is Synthetic And Why

Synthetic field	Why it is needed	How realism is enforced
solar_kw	Converts irradiance into PV power.	Uses PV capacity, performance ratio, temperature/humidity derating, clipping.
load_kw	NASA does not provide site demand.	Morning/evening peaks, weekday/weekend effect, seasonality, cooling load, humidity, noise, events, hard limits.
tariff_inr_kwh	Needed for cost optimization.	Deterministic India-style time-of-use tariff.
battery_soc_pct	Needed for BESS dispatch.	20 to 90 percent SoC safety band.
battery_power_kw	Shows charge/discharge.	Max charge/discharge power and efficiency constraints.
battery_energy_kwh	Converts SoC to stored energy.	Based on 180 kWh capacity.
battery_violation	Safety validation.	Should remain 0.
derived features	Needed for ML.	Hour/day cyclic encodings, lags, rolling averages.

Current Synthetic Validation

Signal	Result
Load min	26.383 kW
Load max	151.913 kW
Load mean	73.084 kW

Signal	Result
Load constraint	24 to 175 kW
Tariff values	2.6, 5.6, 9.2 INR/kWh
Battery capacity	180 kWh
Battery min SoC	20 percent
Battery max SoC	90 percent
Observed min SoC	20 percent
Observed max SoC	90 percent
Battery violations	0

EDA Summary

Detailed generated EDA report:

```
docs/data-eda-report.md
```

EDA plot folder:

```
data/outputs/eda/
```

Key findings:

Finding	Value
Peak load	151.91 kW
95th percentile load	108.57 kW
Solar zero or near-zero hours	48.10 percent
Total simulated load	3,844,827.29 kWh-equivalent
Total PV generation potential	1,275,872.00 kWh-equivalent
Strongest temperature-load lag	0 hours
Correlation at strongest lag	0.6072

Weather Feature Ranges After Cleaning

Feature	Min	Max	Mean
GHI	0.0	1043.15	216.9669

Feature	Min	Max	Mean
DNI	0.0	1100.0	128.9846
Diffuse irradiance	0.0	540.25	104.0249
Temperature	13.43	44.52	26.8362
Wind speed	0.03	14.63	3.2380
Humidity	11.17	100.0	73.7525
Precipitation	0.0	300.0	3.5717

EDA Plot Interpretation

1. Correlation Heatmap

Path:

```
data/outputs/eda/eda_correlation_heatmap.png
```

What it shows:

- Relationships between weather, solar, load, tariff, and battery features.
- Whether solar is strongly tied to irradiance.
- Whether temperature and load are related.

How to use it:

- Justifies feature selection for forecasting.
- Shows that the model has meaningful explanatory variables.

2. Hourly Profile

Path:

```
data/outputs/eda/eda_hourly_profile.png
```

What it shows:

- Average solar/load/tariff behavior by hour.

Expected pattern:

- Solar rises after morning, peaks in daytime, falls at evening.
- Load has morning and evening peaks.
- Tariff is highest in evening.

Why it matters:

- This is the central EMS problem: evening tariff is high when solar is low.
- Battery dispatch is valuable because it can move energy across time.

3. Monthly Profile

Path:

```
data/outputs/eda/eda_monthly_profile.png
```

What it shows:

- Seasonal changes in weather, solar, and load.

Why it matters:

- Summer load increases due to cooling.
- Solar resource varies across months.
- Forecasting and dispatch must handle seasonality.

4. Temperature-Load Lag

Path:

```
data/outputs/eda/eda_temperature_load_lag.png
```

What it shows:

- Correlation between temperature and load at different time lags.

Current finding:

- Strongest correlation is at 0 hours with correlation 0.6072.

Important report note:

- The report text claims an approximately 2.5-hour temperature-load lag.
- Current generated EDA does not support that exact claim.

Defense:

The current synthetic load generator includes immediate cooling load response. Therefore the strongest lag appears at 0 hours. A future version can add thermal inertia to shift peak load response by 2 to 3 hours if exact report alignment is required.

5. Load Duration Curve

Path:

```
data/outputs/eda/eda_load_duration_curve.png
```

What it shows:

- Loads sorted from highest to lowest.
- How often extreme load appears.

Why it matters:

- Helps size battery and grid connection.
- Shows whether high peaks are rare or frequent.
- Supports peak demand risk threshold selection.

6. Signal Distributions

Path:

```
data/outputs/eda/eda_distributions.png
```

What it shows:

- Distribution of key weather and EMS signals.

Why it matters:

- Confirms values are not unrealistic.
- Shows solar has many zero values because night hours exist.
- Shows load distribution stays inside practical range.

7. Recent Battery/Grid Profile

Path:

```
data/outputs/eda/eda_battery_grid_recent.png
```

What it shows:

- Recent battery SoC and grid import behavior.

Why it matters:

- Shows how BESS dispatch affects grid import.
- Helps validate safety and operational behavior.

Data Flow

```
NASA POWER raw CSV
-> cleaning.py
-> cleaned_nasa_power.csv
-> synthetic.py
-> ems_dataset.csv
-> LSTM training / PPO training / metrics / API
-> React dashboard
```

Live mode data flow:

```
Start live stream button
-> POST /live/start
-> live_simulator.py background thread
-> data/runtime/live_telemetry.jsonl
-> recent_window()
-> forecast / optimize / metrics / alerts endpoints
-> dashboard refresh every 1.5 seconds
```

Why Synthetic Data Is Defensible

Synthetic data is defensible because:

- NASA POWER provides weather but not private facility demand or BESS telemetry.
- Facility load profiles are often unavailable due to privacy and hardware access.
- Synthetic load follows real constraints: morning/evening peaks, weekday/weekend variation, seasonal cooling, weather effects, noise, and bounded spikes.
- Battery simulation respects capacity, SoC limits, charge/discharge limits, and efficiency.
- Tariff is deterministic and within India TOU ranges.

Important:

Synthetic data should be presented as a realistic operating layer built on top of real NASA weather data, not as measured building load data.

Best Way To Explain The Dataset In Presentation

Use this statement:

The meteorological base is real NASA POWER hourly data. Since the project does not have access to a private site's smart meter, tariff bill, inverter, and BMS telemetry, the operational layer is generated synthetically but constrained by real microgrid behavior: bounded load, time-of-use tariff, PV physics, battery capacity, efficiency, and SoC safety limits.

Data Limitations

Limitation	Effect	Defense
No real smart meter load	Load is synthetic.	Synthetic load is constrained and realistic.
No real BMS telemetry	Battery behavior is simulated.	Physics constraints are implemented and validated.
No real inverter telemetry	PV output is derived from irradiance.	PV formula uses capacity, PR, and derating.
No real tariff API	Tariff schedule is fixed.	Deterministic TOU tariff matches project requirement.
NASA data is hourly	Limited high-frequency weather detail.	Live simulator provides high-frequency dashboard behavior, while training uses hourly data.

EDA Conclusion

The EDA supports the EMS design:

- Solar is variable and often unavailable at night.
- Load has peaks and weather dependency.
- Evening tariff creates an economic problem.
- Battery dispatch can reduce peak grid import.
- Forecasting is useful because the best battery action depends on future load, solar, and tariff.
- The generated synthetic layer remains within realistic ranges and battery safety constraints.